Guided Lecture Notes

for

Proof, Set Theory, and Logic

MATH 330

Complete packet

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These guided notes accompany the Fall 2026 revised course edition of *Elementary Set Theory: Proof, Set Theory, and Logic*. They are based on Richard P. Millspaugh's original textbook materials and lecture materials prepared and revised by Bryce A. Christopherson.

The blank spaces in the student copy are completed during class. Each blank occupies exactly the same space as the corresponding typed completion in the instructor copy, so the two packets have identical pagination. Definitions, theorem statements, and the surrounding explanations are provided so that class time can focus on examples, proof decisions, and the mathematical steps that connect them.

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Chapter 1

Elementary Logic

1.1 Propositions

Definition 1.1.
A *proposition* is a statement that is either true or false, but not both. The *truth value* of a proposition is true (T) if the sentence is true and false (F) if the sentence is false.

Example 1.2.
Determine whether the following are propositions:

- *Two plus two equals four.*

- *Seven minus three equals twelve.*

- *It will snow in Grand Forks on March 17, 2525.*

- *Go clean your room.*

- *Is it raining outside?*

- *This sentence is false.*

- $x + 3 = 12$.

We will use uppercase letters, frequently P or Q, to stand for variable propositions in much the same way that you might use x or y to represent variable numbers in algebra.

1.2 Connectives and Truth Tables

Many of the propositions we are interested in are made up of more elementary propositions. For example, the proposition S: *today is Tuesday and it's raining* is made up of the two propositions P: *today is Tuesday* and Q: *it's raining*. The word *and* in S is called a connective since it gives us a way to connect two propositions and form another. In this section we will look at several kinds of connectives.

1.2.1 Conjunction

Definition 1.3.
The *conjunction* of P and Q, denoted $P \wedge Q$, is true exactly when both P and Q are true.

We will sometimes keep track of truth values in a *truth table*, where we list the truth value of a proposition in terms of the truth values of its elementary propositions.

Example 1.4.
Construct a truth table for the conjunction $P \wedge Q$.

P	Q	$P \wedge Q$
T	T	
T	F	
F	T	
F	F	

Note that in this case there are four rows in the table. For a proposition involving n distinct elementary propositions, there are 2^n possible truth assignments and hence 2^n rows in its truth table. This makes it cumbersome to construct truth tables for very complicated propositions. Nevertheless, we will find them to be a convenient tool.

1.2.2 Disjunction

We next consider propositions of the form P *or* Q. The word *or* can be ambiguous in English: sometimes it includes the possibility that both conditions hold, while in other contexts only one option is intended. In logic we use *or* inclusively. Thus $P \vee Q$ is true when at least one of P and Q is true.

Definition 1.5.
The *disjunction* of P and Q, denoted $P \vee Q$, is true whenever at least one of P or Q is true.

Example 1.6.

Construct a truth table for the proposition $P \vee Q$.

P	Q	$P \vee Q$
T	T	
T	F	
F	T	
F	F	

1.2.3 Implication

Most mathematical propositions have the form *If* P, *then* Q. Of course, either of P or Q might be propositions made of several other propositions.

Definition 1.7.
The proposition *If* P, *then* Q is called an *implication* and is denoted $P \Rightarrow Q$. It is false exactly when P is true and Q is false.

Example 1.8.

Construct a truth table for $P \Rightarrow Q$.

P	Q	$P \Rightarrow Q$
T	T	
T	F	
F	T	
F	F	

Notice: To say that $P \Rightarrow Q$ is true **does not** mean that P is true or that Q is true. It **does** mean that there is a relationship between the truth values of these two propositions. The implication $P \Rightarrow Q$ is false when P is true and Q is false; it is true when either P is false or Q is true.

There are several other implications related to the implication $P \Rightarrow Q$ that we will occasionally find useful.

Definition 1.9.
For the implication $P \Rightarrow Q$:

- The *converse* is $Q \Rightarrow P$.
- The *contrapositive* is $\neg Q \Rightarrow \neg P$.
- The *inverse* is $\neg P \Rightarrow \neg Q$.

1.2.4 Logical Equivalence

Definition 1.10.
The *biconditional* $P \Leftrightarrow Q$, read "P if and only if Q," is true exactly when P and Q have the same truth value under the given assignment. Two propositional forms P and Q are *logically equivalent* if they have the same truth value under every assignment of truth values to their elementary propositions. Equivalently, $P \Leftrightarrow Q$ is a tautology.

Example 1.11.
Construct a truth table for $P \Leftrightarrow Q$.

P	Q	$P \Leftrightarrow Q$
T	T	
T	F	
F	T	
F	F	

1.2.5 Negation

Definition 1.12.
The *negation* of P, denoted $\neg P$ and sometimes read "not P," is the proposition whose truth value is always opposite that of P.

Negation is a unary connective: it acts on one proposition, whereas conjunction, disjunction, implication, and biconditional are binary connectives.

Example 1.13.
Construct a truth table for P and $\neg P$.

P	$\neg P$
T	
F	

1.2.6 Tautology and Contradiction

Definition 1.14.
A *tautology* is a proposition that is always true, regardless of the truth values of the elementary propositions that make it up.

Definition 1.15.
A *contradiction* is a proposition that is always false.

Example 1.16.

Show the proposition $P \vee \neg P$ is a tautology and that the proposition $P \wedge \neg P$ is a contradiction using a truth table.

P	$\neg P$	$P \vee \neg P$	$P \wedge \neg P$
T			
F			

Example 1.17.

Use a truth table to show that $P \Rightarrow (P \vee Q)$ is a tautology.

P	Q	$P \vee Q$	$P \Rightarrow (P \vee Q)$
T	T		
T	F		
F	T		
F	F		

Example 1.18.

Use a truth table to show that $(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$ is a tautology.

P	Q	$\neg P$	$\neg Q$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$
T	T				
T	F				
F	T				
F	F				

1.3 Predicates, Instantiation, and Quantification

We return our attention now to statements that involve variables, like $x + 3 = 12$. Such a statement is called a *predicate* and we will usually indicate it with $P(x)$ rather than P. Before a predicate takes on a truth value, we need to know something about the variable. We noted before that if $x = 0$, then the statement is false; but if $x = 9$, then the statement is true. What if x represents my desk? In that case the statement is complete nonsense, but you probably feel like there's something a little bit unfair about that choice for x. We see an equation and usually think that the variable must represent a number, right? The first thing you should know about any variable in a predicate is what it can represent. This is called the *universe* or the *domain of discourse*. In mathematics, we will be very explicit about what universe we are talking about.

1.3.1 Instantiation

Suppose the universe for the preceding predicate $P(x)$ is $\mathbb{R}$. That still doesn't turn $P(x)$ into a proposition because, as noted above, $P(x)$ doesn't take on a truth value until we know which real number x indicates. One way to do this is called *instantiation*, which means choosing a particular value for the variable(s). Consider for example the sentence: *Let* $x = 2$, *then* $x + 3 = 12$. In this case the sentence is a proposition with truth value F. The other option is *quantification* of the variable(s). We will discuss two kinds of quantifiers, *universal* and *existential*. You might occasionally encounter other quantifiers, but they are not necessary.

1.3.2 Universal Quantification

Definition 1.19.
A *universal quantifier* indicates that the predicate is true for all instances of the variable in the given universe. It is typically indicated by using the phrase "for every" or "for all," and can be denoted symbolically by $\forall$ in symbolic statements.

The following sentences are universally quantified propositions. Only the second is true.

- *For every real number* x, $x + 3 = 12$.
- *The square of* x *is nonnegative for all real* x.
- *The square of* x *is real for all complex* x.

The third statement is false; for example, $(1 + i)^2 = 2i$, which is not real.

1.3.3 Existential Quantification

Definition 1.20.
An *existential quantifier* indicates that the predicate is true for at least one instance of the variable in the given universe. It is typically indicated by the phrase "for some" or "there exists," and can be denoted by $\exists$ in symbolic statements.

The following sentences are existentially quantified. All three of these are true.

- *There is a real number* x *so that* $x + 3 = 12$.
- *The square of* x *is nonnegative for some real* x.
- *There is a complex number* x *whose square is* -1.

1.3.4 Syllogisms and Venn Diagrams

A *syllogism* is a form of logical argument that deduces a conclusion based on two premises. For example:

All men are mortal. Socrates is a man. Hence Socrates is mortal.

This is a valid syllogism. If we assume that the premises are true, then the conclusion must follow. One tool we can use to help think about syllogisms is a *Venn diagram*, which in its simplest form is a collection of circles that represent the categories in the premises.

Example 1.21.
Construct a Venn diagram to determine if the following syllogism is valid: *All men are mortal. Socrates is a man. Hence Socrates is mortal.*

Example 1.22.

Determine whether the following syllogism is valid:
Some students are freshmen. Pat is a student. Hence Pat is a freshman.

Example 1.23.

Determine whether the following syllogism is valid:
No cats are dogs. Jade is a cat. Hence Jade is not a dog.

Example 1.24.

Determine whether the following syllogism is valid:
Some cats are psychopaths. Jawa is a cat. Hence Jawa is a psychopath.

Example 1.25.

Determine whether the following syllogism is valid:
Some Billywiggles are Bleepzigs. Some Bleepzigs are Jabberwoks. Hence some Billywiggles are Jabberwoks.

1.3.5 Quantification of Several Variables

A predicate can have two or more variables. In order to use the predicate in a proposition, a universe must be declared for each variable and each variable must be instantiated or quantified. We will discuss the various ways a two-variable predicate might be quantified by considering an example.

Example 1.26.

Consider the predicate $\mathrm{P}(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\forall x\, \forall y\, \mathrm{P}(x, y)$

Example 1.27.

Consider the predicate $\mathrm{P}(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\forall y\, \forall x\, \mathrm{P}(x, y)$.

Example 1.28.

Consider the predicate $\mathrm{P}(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\exists x\, \exists y\, \mathrm{P}(x, y)$.

Example 1.29.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\forall x\, \exists y\, P(x, y)$.

Example 1.30.

Consider the predicate $P(x, y)$ given by $x + y = 0$ and assume that the universe for both x and y is the set of real numbers. Determine whether the following proposition is true: $\exists y\, \forall x\, P(x, y)$.

1.4 Negations

The negation of a proposition is the proposition with the opposite truth value. In this section we are going to discuss the negations of compound propositions, which can be an important step in some kinds of proofs. We begin with two theorems that tell how to find the negations of propositions involving connectives.

Theorem 1.31.
Given any propositions P *and* Q, $\neg(P \Rightarrow Q) \Leftrightarrow (P \wedge \neg Q)$.

Proof. Complete the truth table below and compare its final two columns.

P	Q	$\neg Q$	$P \Rightarrow Q$	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T				
T	F				
F	T				
F	F				

Theorem 1.32 (De Morgan's Laws)**.**
Given any propositions P *and* Q*:*

(i) $\neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$

(ii) $\neg(P \vee Q) \Leftrightarrow (\neg P \wedge \neg Q)$

You will prove this theorem in the homework.

Example 1.33.
Negate the proposition $P \Rightarrow (Q \land R)$.

Example 1.34.
Find the negation of $P \land (Q \Rightarrow R)$.

1.4.1 Negating Quantified Statements

The quantifiers $\forall$ and $\exists$ are interchanged by negation. This is the quantified analogue of De Morgan's Laws.

Theorem 1.35 (Quantified Negation). *Let* $P(x)$ *be a predicate on a specified domain. Then*

$$\neg\big(\forall x\, P(x)\big) \iff \exists x\, \neg P(x)$$

and

$$\neg\big(\exists x\, P(x)\big) \iff \forall x\, \neg P(x).$$

Proof.

For nested quantifiers, move the negation inward one quantifier at a time. For example,

$$\neg\big(\forall x\, \exists y\, Q(x,y)\big) \iff \exists x\, \forall y\, \neg Q(x,y).$$

Notice that in $\forall x\, \exists y\, Q(x,y)$ the witness y may depend on x; reversing the quantifiers would assert the existence of one choice of y that works for every x.

Example 1.36.

Find the negation of $(P \vee Q) \wedge (P \vee R)$.

Chapter 2

Logical Arguments

2.1 Rules of Inference

We accept the following two basic rules of thought:

The *Law of Excluded Middle* says that given any proposition P, we may deduce $P \vee \neg P$. In words, either P is true or $\neg P$ is true.

The *Law of non-Contradiction* says that given any proposition P, we may deduce $\neg(P \wedge \neg P)$. In words, P and $\neg P$ cannot both be true.

A *rule of inference* is a logical rule consisting of a hypothesis and a conclusion. Whenever the hypothesis is true, the conclusion must also be true. For our purposes this means that once we have deduced all of the hypotheses, we may also deduce the conclusion. Rules of inference are closely related to tautologies. In fact, they are essentially the same thing. While every tautology gives rise to a rule of inference, in practice the seven listed in the table below will usually suffice.

Table 2.1: Rules of Inference

Name	**Hypotheses**	**Conclusion**
Modus ponens	P and $P \Rightarrow Q$	Q
Modus tollens	$\neg Q$ and $P \Rightarrow Q$	$\neg P$
Hypothetical syllogism	$P \Rightarrow Q$ and $Q \Rightarrow R$	$P \Rightarrow R$
Addition	P	$P \vee Q$
Simplification	$P \wedge Q$	P
Conjunction	P and Q	$P \wedge Q$
Disjunctive syllogism	$P \vee Q$ and $\neg P$	Q

The Rule of Addition, for example, comes from the tautology $P \Rightarrow (P \vee Q)$. Since the implication is always true, knowing that the hypothesis is true will tell us that the conclusion must also be true. We can verify each of the Rules of Inference in Table 2.1 by constructing a truth table for the associated tautology.

2.1.1 Propositional Arguments

Each of our Rules of Inference is actually a *propositional argument*, where accepting the hypotheses forces us to accept the conclusion. We use the following notation for propositional arguments, where the propositions above the line are hypotheses and the proposition(s) below the line are the conclusion(s). We precede the conclusion(s) with the symbol $\therefore$, which is read "therefore."

Example 2.1.

Here is the rule of modus tollens as a propositional argument:

$$\begin{array}{c} \neg Q \\ P \Rightarrow Q \\ \hline \therefore \neg P \end{array}$$

Example 2.2.

Express the rule of modus ponens as a propositional argument.

Example 2.3.

Express the rule of hypothetical syllogism as a propositional argument.

Definition 2.4.
A propositional argument is said to be *valid* if accepting the hypotheses forces us to accept the conclusion(s).

How do we determine whether or not a propositional argument is valid? One option is to construct a truth table, but that can be very tedious. Our preferred option is to construct a logical argument which begins by assuming each of the hypotheses is true and deduces consequences using our rules of inference. Each deduced proposition is a step in our proof that can be justified as either a hypothesis, a tautology, a consequence of previous steps and a rule of inference, or a proposition that is logically equivalent to a previous step. The proof is complete when we have deduced the desired conclusion.

Example 2.5.
Prove that the following argument is valid:

$$\begin{array}{c} \mathrm{P} \\ \mathrm{P} \Rightarrow \mathrm{Q} \\ (\mathrm{Q} \vee \mathrm{R}) \Rightarrow \mathrm{S} \\ \hline \therefore \mathrm{S} \end{array}$$

Example 2.6.

Prove that the following argument is valid:

$$\begin{array}{c} P \Rightarrow (Q \wedge \neg Q) \\ \hline \therefore \neg P \end{array}$$

The argument in Example 2.6 says that if a proposition P implies a contradiction, then ¬P must be true. This result is the basis for proofs by contradiction.

2.1.2 Arguments with Quantifiers

Arguments involving quantified propositions require rules for instantiation and generalization. The *rules for instantiation* let us deduce statements about particular elements of the domain from quantified statements. The *rules for generalization* tell us when deductions about elements of the domain justify quantified statements. We give these rules in Table 2.2.

Table 2.2: Rules of Quantification

Name	**Rule**
Universal Instantiation	$\forall x\ P(x) \therefore P(c)$ whenever c is in the domain of x
Existential Instantiation	$\exists x\ P(x) \therefore P(c)$ for some particular but unspecified c in the domain of x
Universal Generalization	$P(c)$ for *arbitrary* c in the domain of $x \therefore \forall x\ P(x)$
Existential Generalization	$P(c)$ for *some* c in the domain of $x \therefore \exists x\ P(x)$

The restrictions in this table matter. In existential instantiation, the letter c denotes one particular but unspecified value whose existence is guaranteed by the existential statement. We do not get to choose which value it is, and we may not assume anything about it beyond what the existential statement tells us. The letter c should therefore not already be in use elsewhere in the argument. In universal generalization, by contrast, c represents an arbitrary element of the domain.

Example 2.7.

Prove that the following argument is valid:

$$\begin{array}{c} \forall x\ (\mathrm{P}(x) \land \mathrm{Q}(x)) \\ \exists x\ (\mathrm{P}(x) \Rightarrow \mathrm{R}(x)) \\ \hline \therefore\ \exists x\ (\mathrm{Q}(x) \land \mathrm{R}(x)) \end{array}$$

2.2 Axioms and Formal Mathematical Theories

The rules of inference developed above tell us which deductive steps are valid because of their logical form. To develop a particular area of mathematics, however, we also need starting statements about the objects we intend to study. A *formal mathematical theory* consists of a specified language, logical principles and rules of inference, and a collection of *axioms*. An axiom is a statement assumed throughout the theory rather than proved from earlier statements in that theory.

Axioms should be distinguished from hypotheses and definitions. An axiom is a standing assumption of the entire theory, while a hypothesis is usually a temporary assumption made within a particular argument. A definition assigns meaning to a new term or symbol; it does not, by itself, guarantee that an object having the stated properties exists.

A *theorem* is a statement that can be deduced from the axioms using the accepted rules of inference. For example, if P and $P \Rightarrow Q$ are axioms or previously proved theorems, then modus ponens allows us to deduce Q. Once proved, Q may itself be used in later arguments. In a completely formal proof, every such step is written down and justified.

Calling a statement an axiom does not mean that it is self-evident or true in every possible mathematical setting. It describes the role that the statement plays within a particular theory. Different choices of axioms can lead to different theories and therefore to different collections of theorems.

Ordinary mathematical proofs do not include every elementary formal step. Instead, we write in mathematical English, make the important deductions explicit, and suppress routine applications of logic. Such proofs are nevertheless intended to be formalizable: in principle, the omitted steps could be supplied within an appropriate axiomatic theory. In the next chapter we will briefly examine ZFC, the standard axiomatic framework underlying most ordinary work with sets.

2.3 Proving Mathematical Theorems

Most mathematical statements are implications of the form $P \Rightarrow Q$, where P or Q might be compound propositions. In this section we will look at some methods for proving implications. Note that we are using some definitions and results from other areas of mathematics that we have not explicitly stated. We will not allow ourselves to make those kinds of assumptions once we begin our study of set theory.

2.3.1 Direct Proofs

A direct proof of the implication $P \Rightarrow Q$ assumes that the hypothesis P is true, then uses definitions, previously proved theorems, and the rules of inference to deduce that the conclusion Q must also be true.

Example 2.8.

Prove the following theorem:

Theorem 2.9.
If n is an even integer, then n^2 is even.

Proof.

Although the two-column proof above is valid, it is not written in a format that mathematicians usually find pleasing. We prefer proofs that are written using complete sentences, paragraphs, and correct grammar and punctuation. Here is a better presentation for the previous proof.

Example 2.10.
Prove Theorem 2.9 in plain language.

Proof.

Example 2.11.
Prove the following theorem:

Theorem 2.12.
The square of an odd integer is odd.

Proof.

2.3.2 Proving the Contrapositive

As you have shown in a homework exercise, the contrapositive of an implication is logically equivalent to the implication. Sometimes it is easier to prove the contrapositive of the implication we are interested in, as we will show here.

Example 2.13.
Prove the following theorem:

Theorem 2.14.
If n is an integer and n^2 is even, then n is even.

Proof.

Example 2.15.
Prove the following theorem:

Theorem 2.16. *If n is an integer and n^2 is odd, then n is odd.*

Proof.

2.3.3 Proof by Contradiction

A proof by contradiction, or *reductio ad absurdum*, begins by assuming that the result we are trying to prove is false, then deriving a contradiction from that assumption. It is easy to confuse this with a direct proof of the contrapositive, which makes different beginning assumptions. To prove the implication $P \Rightarrow Q$ by proving the contrapositive, you assume $\neg Q$ and use that to prove $\neg P$. A proof by contradiction assumes $P \wedge \neg Q$ and derives a contradiction $R \wedge \neg R$. Here is a classic example of a proof by contradiction.

Example 2.17.

Prove the following theorem:

> **Theorem 2.18.**
> *There is no rational number x such that $x^2 = 2$.*

Proof.

2.3.4 Proof by Cases

To prove the implication $P \Rightarrow Q$ by cases, we make use of the following argument:

$$\begin{array}{c} P \Rightarrow (R \vee S) \\ R \Rightarrow Q \\ S \Rightarrow Q \\ \hline \therefore P \Rightarrow Q \end{array}$$

You will be asked to show that this argument is valid in the homework.

Here is a very simple example of a proof by cases. We make use of some of the order properties of the real numbers. In particular, we know that multiplying both sides of an inequality by a positive number preserves the inequality, and multiplying both sides of an inequality by a negative number reverses the inequality.

Example 2.19.

Prove the following theorem:

Theorem 2.20.
If x is a real number, then $x^2 \geq 0$.

Proof.

2.4 Proving Quantified Statements

2.4.1 Universally Quantified Statements

On the simplest level, a universally quantified statement has the form $\forall x\, P(x)$. To prove it, let x be an arbitrary element of the required domain and prove $P(x)$ without using any special property of x beyond its membership in that domain. It is frequently helpful to think of such a statement as an implication whose hypothesis is that x belongs to the required domain and whose conclusion is $P(x)$. Since universally quantified statements can be viewed as implications, we can use the same proof techniques.

2.4.2 Existentially Quantified Statements

Existentially quantified statements are significantly different than implications and universally quantified statements. To prove the statement $\exists x, P(x)$ we must show that there is at least one x in the domain that satisfies $P(x)$. One way to do this is simply to find a single particular example.

Example 2.21.

Prove the following theorem:

Theorem 2.22.
There is a positive integer N *such that* N, $N+1$, $N+2$, $N+3$, *and* $N+4$ *are all composite (not prime).*

Proof.

There are some instances where it is easier to show that some x satisfies $P(x)$ without actually finding a particular x that works. Here is an elementary example that works by demonstrating that one of two possible choices must work, without determining which it is.

Example 2.23.

Prove the following theorem:

Theorem 2.24.
There exist irrational numbers a and b so that a^b is rational.

Proof.

2.4.3 Counterexamples

An example showing that a universally quantified statement is false is called a *counterexample* to that statement.

Example 2.25.

Disprove the following proposition:

Conjecture 2.26.
Every integer is even.

2.5 Mathematical Induction

There is another type of proof that is frequently used to prove statements about the natural numbers, i.e., the numbers $1, 2, 3, \ldots$.

Theorem 2.27 (The Principle of Mathematical Induction).
Let $\mathrm{P}(n)$ *be a statement about the natural number* n. *If* $\mathrm{P}(1)$ *is true and if* $\mathrm{P}(k) \implies \mathrm{P}(k+1)$ *for every natural number* k, *then* $\mathrm{P}(n)$ *is true for every natural number* n.

All proofs by induction use the following basic outline. To prove $\mathrm{P}(n)$ for all natural numbers n,

1. Base Case: Show that $\mathrm{P}(1)$ is true.
2. Inductive Step: Prove the implication $\mathrm{P}(k) \implies \mathrm{P}(k+1)$ for all $k \in \mathbb{N}$. Note: in proving that $\mathrm{P}(k) \implies \mathrm{P}(k+1)$, the assumption that $\mathrm{P}(k)$ is true is often referred to as the inductive hypothesis.
3. It is considered good form to clearly state that the statement $\mathrm{P}(n)$ is now true for all natural numbers n by induction.

Example 2.28.

Prove the following theorem:

Theorem 2.29.
For each natural number n, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Proof.

Chapter 3

Set Theory

3.1 What is a Set?

3.1.1 Naive Set Theory

Early work with sets often used *unrestricted comprehension*: any condition was assumed to determine a set of all objects satisfying that condition. Russell's paradox shows that this principle is inconsistent. Modern axiomatic set theories replace it with restricted principles, such as forming a subset of an already specified set. In this course we use informal set language, but every set-builder description should have a stated ambient set.

3.1.2 Russell's Paradox

Example 3.1.

The Barber's Paradox. A certain small town has one barber, who is a man and shaves exactly those men in town who do not shave themselves. Does the barber shave himself?

Russell considered this sort of thing more formally by constructing the (purported) set

$$\mathrm{R} = \{x \mid x \notin x\}.$$

If $\mathrm{R} \in \mathrm{R}$, then its defining condition gives $\mathrm{R} \notin \mathrm{R}$; if $\mathrm{R} \notin \mathrm{R}$, then its defining condition gives $\mathrm{R} \in \mathrm{R}$. The lesson is not merely that self-membership should be forbidden. Rather, the collection of all objects satisfying an arbitrary condition need not be a set. Standard axiomatic systems avoid the contradiction by restricting comprehension, as described below.

3.1.3 A Glimpse of ZFC

The most commonly used formal foundation for ordinary mathematics is *Zermelo–Fraenkel set theory with the Axiom of Choice*, usually abbreviated *ZFC*. It is not the only possible foundation, but it is the standard one. ZFC uses first-order logic with equality and one basic relation, membership $\in$. Every variable in its formal language ranges over sets. Numbers, ordered pairs, functions, and other mathematical objects can all be represented using sets, although we will continue to treat them as the familiar objects that they are. For example, one common set-theoretic representation of an ordered pair is

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

The *Axiom of Extensionality* formalizes our criterion for equality of sets:

$$\forall A\, \forall B \left[\left(\forall x\, (x \in A \Leftrightarrow x \in B) \right) \Rightarrow A = B \right].$$

In words, if two sets have exactly the same elements, then they are equal. Consequently, a set is completely determined by its elements.

The principle that most directly addresses Russell's paradox is the *Separation schema*. For every property $\varphi(x)$ that can be expressed in the language of set theory, Separation supplies an axiom of the form

$$\forall A\, \exists B\, \forall x \left[x \in B \Leftrightarrow \left(x \in A \wedge \varphi(x) \right) \right].$$

Thus, from an already existing set A, we may form the subset

$$B = \{x \in A \mid \varphi(x)\}.$$

The word *schema* means that this is actually a pattern giving one axiom for each allowable formula φ. Separation does not say that $\{x \mid \varphi(x)\}$ exists without an ambient set. In particular, it does not provide Russell's purported set $\{x \mid x \notin x\}$.

The other axioms guarantee the existence of the sets needed for familiar constructions. Pairing gives sets such as $\{a, b\}$; Union and Power Set justify the formation of unions and power sets; and Infinity guarantees the existence of an infinite set. The Replacement schema allows the outputs of a definable functional rule, when applied to the elements of a set, to be collected into another set. Foundation rules out membership loops such as $x \in x$. Finally, the Axiom of Choice says that, given a set of nonempty sets, there is a function choosing one element from each of them. ZF consists of the preceding set-theoretic principles without Choice; ZFC means ZF together with Choice. Equivalent presentations sometimes package a few of these principles differently.

These axioms are what justify the constructions that we use informally. For example, Pairing guarantees that $\{A, B\}$ exists, and the Union axiom then guarantees the existence of $A \cup B$. The Power Set axiom guarantees the existence of $\mathcal{P}(A)$. A definition describes the intended set, while the axioms ensure that such a set actually exists.

We will not derive our results directly from the axioms of ZFC. The purpose of this discussion is only to show the formal framework underneath our informal work. In particular, ZFC has no set containing every set. When we later use a universal set U, it will mean a fixed ambient set for the problem under consideration, not a set containing all mathematical objects.

3.2 Elements and Subsets

3.2.1 Elements

We will think of a set as a collection of objects, called its *elements*. These objects can be any objects we are interested in discussing. Any roster or description must determine membership unambiguously. In particular, a set-builder description should specify an ambient set and a condition that determines membership. We consider two sets equal if they contain exactly the same elements. An object is either an element of a set or it is not; elements have no order and repetition does not create a new element.

One way to indicate a set is to simply list all of the elements between set braces; i.e., between the symbols $\{$ and $\}$. The set of positive integers less than 3 is $\{1, 2\}$. We will find it useful to have a special notation for the set which has no elements. To this end, we let the symbol $\varnothing$ denote the set $\{\}$ (i.e. $\varnothing = \{\}$).

Example 3.2.

Example 3.3.

We indicate that the object x is an element of the set A by writing $x \in A$. We write $x \notin A$ if x is not an element of A.

Listing all elements works well for a small finite set. For larger sets we use *set-builder notation*

$$\{x \in U \mid P(x)\},$$

read "the set of all x in U such that $P(x)$." The ambient set U is part of the definition. For example, $\{x \in \mathbb{R} \mid x^2 - 2 = 0\}$ is a two-element set, whereas $\{x \in \mathbb{Q} \mid x^2 - 2 = 0\} = \varnothing$.

There are also a few sets that are useful enough to deserve their own special notation. In particular, $\mathbb{N}$ denotes the set of natural numbers, $\mathbb{Z}$ the set of integers, $\mathbb{Q}$ the set of rational numbers, $\mathbb{R}$ the set of real numbers, and $\mathbb{C}$ the set of complex numbers.

3.2.2 Subsets

Definition 3.4.
Given two sets A and B, we say that A is a *subset* of B, or that A is contained in B, if every element of A is also an element of B. We write $A \subseteq B$.

Since every natural number is also an integer and every integer is a real number, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{R}$.

Example 3.5.

Given any set A, it is certainly true that every element of A is an element of A. In other words, every set is a subset of itself. Note also that the empty set is a subset of every set A since there are no elements of the empty set which are not in A.

Definition 3.6.
A subset B of A is a *proper subset* of A if $B \neq A$. We write $B \subsetneq A$. In particular, $\varnothing \subsetneq A$ whenever A is nonempty.

Theorem 3.7.
If $A \subseteq B$ *and* $B \subseteq C$, *then* $A \subseteq C$.

Proof.

3.2.3 Universal Sets

In many instances, all of the sets we may be interested in are subsets of some particular set U. In this case we say that U is a *universal set*, or that U is the *universe*. For example, all of the functions we study in single variable calculus have domains and ranges that are subsets of the universal set $\mathbb{R}$.

3.2.4 Equality of Sets

We say that two sets are equal if they contain exactly the same elements. In other words, $A = B$ means that every element of A is an element of B and every element of B is an element of A. A good test for this is the following one:

Theorem 3.8.
Given any two sets A *and* B, $A = B$ *if and only if* $A \subseteq B$ *and* $B \subseteq A$.

This theorem will become one of our most valuable tools for showing that two sets are equal.

3.3 Operations on Sets

3.3.1 Union and Intersection

Definition 3.9.
The *union* of two sets A and B is the set:

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

Definition 3.10.
The *intersection* of A and B is the set:

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

Example 3.11.

Theorem 3.12.
Let A, B, and C be any sets in the universal set U. Then:

1. *(Commutative Laws)*
   (a) $A \cup B = B \cup A$
   (b) $A \cap B = B \cap A$
2. *(Associative Laws)*
   (a) $A \cup (B \cup C) = (A \cup B) \cup C$
   (b) $A \cap (B \cap C) = (A \cap B) \cap C$
3. *(Distributive Laws)*
   (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. *(Idempotence)*
   (a) $A \cup A = A$
   (b) $A \cap A = A$
5. *(Identities)*
   (a) $A \cup \varnothing = A$
   (b) $A \cap U = A$

Proof.

3.3.2 Complements

Definition 3.13.
The *relative complement* of B in A (also called the set difference) is the set:

$$A \setminus B = \{a \mid a \in A \text{ and } a \notin B\}$$

In a given universe U, we may also define the *complement* of a set A to be the set:

$$A' = U \setminus A$$

Theorem 3.14 (De Morgan's Laws).
Let A, B, *and* C *be sets. Then:*

(i) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$, *and*

(ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Proofs.

3.3.3 Cartesian Products

Definition 3.15.
The *(Cartesian) product* of two sets A and B is the set:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\},$$

where (a, b) denotes an ordered pair, not an interval.

Ordered pairs satisfy $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$.

Note that $A \times B$ is simply the set of all ordered pairs with first coordinates in A and second coordinates in B. For example, the Cartesian plane used in Calculus is the set $\mathbb{R} \times \mathbb{R}$.

Theorem 3.16.
For any sets A, B, *and* C*:*

(i) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

(ii) $(A \cap B) \times C = (A \times C) \cap (B \times C)$

(iii) $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$

Proof.

The products $A \times B$ and $B \times A$ need not be equal because their coordinates play different roles. They are equal in special cases, for example when $A = B$ or when at least one factor is empty. The proof of the following theorem involves making obvious changes to the previous proof and checking that all of the details still work.

Theorem 3.17.
For any sets A, B, *and* C*:*

(i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(iii) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$

It may be tempting to assume that we could combine these two theorems somehow and obtain statements like $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$. This statement is generally false, however.

Example 3.18.

3.4 Collections of Sets

Some of the structures used in pure mathematics require the use of sets whose elements are other sets. It is customary, though not necessary, to refer to these kinds of sets as collections of sets.

3.4.1 The Power Set of a Set

Definition 3.19.
Let A be any set. The *power set* of A is $\mathcal{P}(A) = \{B \mid B \subseteq A\}$.

In words, the power set of A is the set whose elements are the subsets of A. Note that for any set A we have $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$, so $\mathcal{P}(A)$ is nonempty for every set A. The power set of A is frequently denoted 2^A.

Theorem 3.20.
For any sets A *and* B, $A \subseteq B$ *if and only if* $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Proof.

Theorem 3.21.
For any sets A *and* B, $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Proof.

Chapter 4

Relations

Introduction

The first mathematical relation most students become aware of is the standard order on the natural numbers, which is later extended to the integers and finally the reals. A relation is a much more general concept, and by now you have probably worked with several others, perhaps without thinking about them as relations. In this chapter we define what we mean by a relation between sets, then define an equivalence relation on a set and use that idea to develop the set of rational numbers.

4.1 Relations

Definition 4.1.
Given two sets A and B, a *relation* from A to B is a subset of $A \times B$. The relations we will be most interested in are usually from a set A to itself, in which case we say that the relation is a relation on A. If S is a relation, we frequently use the notation xSy to indicate that the ordered pair (x, y) is in S.

Let's look at some examples of relations.

Example 4.2.

Example 4.3.

Example 4.4.

Definition 4.5.
Let A be a set and let $\sim$ be a relation on A. We say that $\sim$ is:

1. *reflexive* if $a \sim a$ for every $a \in A$;
2. *symmetric* if $a \sim b$ implies $b \sim a$ for every $a, b \in A$;
3. *antisymmetric* if $a \sim b$ and $b \sim a$ imply $a = b$ for every $a, b \in A$.
4. *transitive* if $a \sim b$ and $b \sim c$ imply $a \sim c$ for every $a, b, c \in A$.

Note that symmetry and antisymmetry are not logical opposites, though the names may lead you to believe otherwise. It is possible for a relation to satisfy both of these properties, or to satisfy neither of them.

Example 4.6.

4.2 Partial Orders

Definition 4.7. A relation $\preceq$ on a set A is a *partial order* if it is reflexive, antisymmetric, and transitive. The pair $(A, \preceq)$ is a *partially ordered set*, or *poset*.

Two elements $a, b \in A$ are *comparable* if $a \preceq b$ or $b \preceq a$. A partial order in which every pair of elements is comparable is called a *total order*.

Example 4.8.

Example 4.9.

4.3 Equivalence Relations

Consider the following question: Are $\pi/4$ and $25\pi/4$ measurements of the same angle? It's certainly true that when we place these angles in standard position they have the same terminal side, which means that we can treat them as the same much of the time. On the other hand, if we think of the angle as a rotation, these two angles are certainly different. If you're playing pin the tail on the donkey, being spun through an angle of $25\pi/4$ is going to make you dizzier than being spun through an angle of $\pi/4$. Intuitively, the terminal side of the angle in this setting tells us what direction we end up pointing in. The angle tells us how far we rotated to end up pointing in that direction. When we are only concerned with the terminal side of an angle, how do we tell when two numbers are measures for *equivalent* angles? You probably learned in a trigonometry or precalculus class that two measurements represent equivalent angles when they differ by an integer multiple of 2π. In other words, two angles α and β are said to be *equivalent* when there is an integer n such that $\alpha - \beta = 2\pi n$. In the following example we consider this relation between real numbers.

Example 4.10.

Definition 4.11.
A relation on a set A that is reflexive, symmetric, and transitive is said to be an *equivalence relation* on A.

4.3.1 Equivalence Classes

Definition 4.12.
Let A be a set and let $\sim$ be an equivalence relation on A. For any $a \in A$ the set $[a] = \{b \in A \mid b \sim a\}$ is called the *equivalence class* of a.

Given a set A and an equivalence relation on A, the set of equivalence classes forms a *partition* of A. In other words, the equivalence classes are nonempty subsets of A, every element of A belongs to an equivalence class, and no two distinct equivalence classes intersect. We formalize this in the following theorem.

Theorem 4.13.
Let A *be a nonempty set and* $\sim$ *an equivalence relation on* A*. For each* $a \in A$*, let*

$$[a] = \{b \in A \mid b \sim a\}.$$

(i) For each $a \in A$*,* $a \in [a]$*.*

(ii) If $[a] \neq [b]$*, then* $[a] \cap [b] = \emptyset$*.*

Proof.

Example 4.14.

In the previous example, note that we had several ways to refer to each equivalence class. For example $[1] = [16]$, so we may as well have used $[16]$ as a name for this equivalence class. In this context the numbers 1 and 16 (or any other member of the class) are called *representatives* of this equivalence class, and any representative can be used to name the equivalence class.

4.4 The Rational Numbers

In this section we use equivalence classes to construct the rational numbers. Assume that $\mathbb{N}$ and $\mathbb{Z}$ have their usual properties and recall that $0 \notin \mathbb{N}$ in this text. Begin with

$$\mathbb{F} = \mathbb{Z} \times \mathbb{N}.$$

Its elements are ordered pairs (a, b), not yet numerical quotients. We use a/b as a convenient *formal symbol* for the pair (a, b); the condition $b \in \mathbb{N}$ guarantees that its second coordinate is nonzero.

Note that a fraction is not quite the same thing as a rational number, for example $1/2$ and $3/6$ are different fractions that represent the same rational number. We define an equivalence relation $\cong$ on $\mathbb{F}$ that clarifies when two fractions represent the same rational as follows:

Definition 4.15.
Given two fractions a/b and c/d in $\mathbb{F}$, we say that $a/b \cong c/d$ if $ad = bc$.

Theorem 4.16.
The relation $\cong$ is an equivalence relation on $\mathbb{F}$.

Proof.

We now define the set of rationals to be the set of equivalence classes of fractions. In this construction the two fractions 1/2 and 3/6 do indeed represent the same rational number since they are representatives of the same equivalence class. This construction might be a bit unsatisfying in the sense that it tells us nothing about how we might add or multiply rational numbers. Even if we know how to add fractions, how would we add two equivalence classes of fractions? One possibility is that we might add two equivalence classes by adding their representatives, so for example we might try:

$$\left[\frac{1}{2}\right] \oplus \left[\frac{2}{3}\right] = \left[\frac{1(3) + 2(2)}{2(3)}\right] = \left[\frac{5}{6}\right]$$

There is a potential problem with this, though. There are infinitely many choices of fractions representing each rational number. Are we sure that we arrive at the same result for all of those possible choices? The next theorem says that we do.

Theorem 4.17.
Suppose that $a/b \cong x/y$ and $c/d \cong w/z$, where a/b, c/d, x/y, and w/z are all in $\mathbb{F}$, then:

$$\frac{ad+bc}{bd} \cong \frac{xz+yw}{yz}.$$

Proof.

The proof of the following theorem is left as an exercise.

Theorem 4.18.
Suppose that $a/b \cong x/y$ and $c/d \cong w/z$, where a/b, c/d, x/y, and w/z are all in $\mathbb{F}$, then:

$$\frac{ac}{bd} \cong \frac{xw}{yz}.$$

Theorem 4.17 and Theorem 4.18 allow us to make the following definitions, where we may choose any representative from each equivalence class.

Definition 4.19.
Let $[a/b]$ and $[c/d]$ be rational numbers, then we define addition and multiplication by:

$$\left[\frac{a}{b}\right] \oplus \left[\frac{c}{d}\right] = \left[\frac{ad+bc}{bd}\right] \quad \text{and} \quad \left[\frac{a}{b}\right] \otimes \left[\frac{c}{d}\right] = \left[\frac{ac}{bd}\right].$$

Chapter 5

Functions

Introduction

As anybody who has taken a calculus class knows, functions are important in mathematics. In this chapter we will define a function between two sets as a special type of relation between those sets. This set-theoretic definition emerged from nineteenth- and early twentieth-century efforts to make the notions of function, relation, and set mathematically precise. These efforts became especially important once paradoxes exposed limitations in informal set reasoning.

5.1 Definition

You probably think of functions in several ways based on what you have seen in previous courses. Functions can be rules for computing things, some people think of them as machines (plug in one number and get out another), and of course we all know that a graph is only the graph of a function if it passes the vertical line test. Most of these ideas match the way that mathematicians thought of functions (and the way they worked with them) at some time or another. It wasn't until the late 19th century that the concept of a function was defined in terms of sets. We present such a definition here:

Definition 5.1.
Let A and B be sets. A *function* f from A to B is a relation from A to B with the additional property that for each $a \in A$ there is exactly one ordered pair (a, b) in f having a as a first coordinate. In this case, the set A is called the *domain* of f and the set B is called the *codomain* of f. If f is a function from A to B, we denote this by writing $f : A \to B$.

Do not confuse the codomain of a function with its range. The codomain of a function is the set that second coordinates must come from. The range is the subset of the codomain containing all of those second coordinates. Sometimes those sets are the same, but sometimes they are not.

Note that our definition requires that every element of the domain be the first coordinate of one, and only one, ordered pair in a function. It does not, however, require that each element of the codomain be a second coordinate, or that it be the second coordinate of only one ordered pair. Satisfying these requirements would make our function surjective or injective, respectively. We discuss these kinds of functions in Section 5.2.

Before going any further, let's consider a couple of examples.

Example 5.2.

The functions in the previous example are obviously constructed to fit the definition, but may not strike you as the kinds of things you think of as functions. What about the kinds of functions we are used to?

Example 5.3.

It probably seems unnatural to you to write the squaring function $f : \mathbb{R} \to \mathbb{R}$ as we did in Example 5.3. Wouldn't it be easier to write this function as $f(x) = x^2$, the same way we did in algebra or calculus classes? Once we know that f is a function, we can use this notation.

Suppose that $f : A \to B$ is a function. If $a \in A$ and $(a, b) \in f$, we say that b is the *value* of the function f at a and denote this by writing $b = f(a)$.

Now the notation $f(x) = x^2$ indicates that for each $x \in \mathbb{R}$, x^2 is the value of the function at x. Please be careful to distinguish between the name of the function f and an arbitrary value of the function $f(x)$.

Example 5.4.

5.1.1 Images and Preimages

Definition 5.5. Let $f : A \to B$ be a function.

(i) If $S \subseteq A$, the *image* of S under f is

$$f[S] = \{f(s) \mid s \in S\} \subseteq B.$$

In particular, the range of f is $f[A]$.

(ii) If $T \subseteq B$, the *preimage* of T under f is

$$f^{-1}[T] = \{a \in A \mid f(a) \in T\}.$$

The notation $f^{-1}[T]$ denotes a set and is meaningful for *every* function f, whether or not f has an inverse function. It should not be confused with the value $f^{-1}(y)$ of an inverse function. For example, if $f : \mathbb{R} \to \mathbb{R}$ is $f(x) = x^2$, then

$$f^{-1}[\{4\}] = \{-2, 2\},$$

even though f has no inverse function from $\mathbb{R}$ to $\mathbb{R}$.

Images and preimages interact differently with set operations. For $S_1, S_2 \subseteq A$ and $T_1, T_2 \subseteq B$,

$$\begin{aligned} f[S_1 \cup S_2] &= f[S_1] \cup f[S_2], \\ f^{-1}[T_1 \cup T_2] &= f^{-1}[T_1] \cup f^{-1}[T_2], \\ f^{-1}[T_1 \cap T_2] &= f^{-1}[T_1] \cap f^{-1}[T_2], \\ f^{-1}[B \setminus T_1] &= A \setminus f^{-1}[T_1]. \end{aligned}$$

In general only $f[S_1 \cap S_2] \subseteq f[S_1] \cap f[S_2]$; equality holds whenever f is injective.

5.1.2 Binary Operations

Standard addition and multiplication on $\mathbb{R}$ are both examples of functions used to compute a single real number from two given real numbers. This kind of function is important enough to deserve its own name.

Definition 5.6.
A *binary operation* on a set X is a function $* : X \times X \to X$. If $*$ is a binary operation on X, we usually use the notation $x * y$ to denote the value $*(x, y)$.

Example 5.7.

Example 5.8.

Definition 5.9. We say that a binary operation $*$ on a set X is:

- *commutative* if $x * y = y * x$ for all $x, y \in X$.
- *associative* if $(x * y) * z = x * (y * z)$ for all $x, y, z \in X$.

Addition and multiplication on $\mathbb{Z}$ (or on $\mathbb{N}$ or $\mathbb{R}$ for that matter) are commutative and associative, which you have probably known since your first algebra class. Subtraction on $\mathbb{Z}$ is not commutative since, for example, $4 - 2 \neq 2 - 4$. Subtraction on $\mathbb{Z}$ also fails to be associative since, for example, $(5 - 1) - 3 = 1 \neq 7 = 5 - (1 - 3)$.

Example 5.10.

5.2 Injective, Surjective, and Bijective Functions

Definition 5.11.
A function $f : X \to Y$ is said to be:

- *injective* or *one-to-one* if for all $x, y \in X$, if $f(x) = f(y)$, then $x = y$.
- *surjective* or *onto* if for each $y \in Y$, there is an $x \in X$ such that $f(x) = y$.
- *bijective* or *a one-to-one correspondence* if f is both injective and surjective.

Example 5.12. Classify the functions from Example 5.4 as injective, surjective, both, or neither, under the assumptions stated below.

(i) Let A be nonempty and consider the identity function $i : A \to A$.

(ii) Assume that A and B each have more than one element and let $k : A \to B$ be the constant function $k(a) = b_0$.

(iii) Assume that A and B each have more than one element and consider the coordinate projections $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$.

(iv) Consider the characteristic function $\chi_A : \mathbb{R} \to \{0, 1\}$.

5.3 Compositions of Functions

Definition 5.13.
Let $f : X \to Y$ and $g : Y \to Z$ be functions. The *composition* is the function from X to Z defined by $(g \circ f)(x) = g(f(x))$. In other words, to find $(g \circ f)(x)$, first find $f(x)$, then plug the result into the function g. In terms of ordered pairs, the composition is

$$g \circ f = \{(x, z) \mid (x, y) \in f \text{ and } (y, z) \in g \text{ for some } y \in Y\}.$$

Theorem 5.14. *Let $f : X \to Y$ and $g : Y \to Z$ be functions.*

(i) If f and g are injective, then $g \circ f : X \to Z$ is injective.

(ii) If f and g are surjective, then $g \circ f : X \to Z$ is surjective.

Proof.

Combining the two parts of Theorem 5.14, we have the following:

Corollary 5.15.
If $f : X \to Y$ and $g : Y \to Z$ are bijective functions, then $g \circ f : X \to Z$ is bijective.

It is natural to ask whether or not the converses of the statements in Theorem 5.14 are true. We consider the converse of Theorem 5.14 *(i)* in the following example and theorem.

Example 5.16.

This example shows that it is possible for $g \circ f$ to be injective when g is not. The next result says that if $g \circ f$ is injective, then it does follow that f is injective.

Theorem 5.17.
Let $f : X \to Y$ and $g : Y \to Z$ be functions such that $g \circ f : X \to Z$ is injective. Then $f : X \to Y$ must be injective.

Proof.

We leave the proof of the corresponding result for surjective functions as an exercise.

Theorem 5.18.
Let $f : X \to Y$ and $g : Y \to Z$ be functions such that $g \circ f : X \to Z$ is surjective. Then $g : Y \to Z$ must be surjective.

5.3.1 Inverse Relations and Inverse Functions

Definition 5.19.
If $f : X \to Y$ is a function, the *inverse relation* of f is the relation g from Y to X given by

$$g = \{(y, x) \mid (x, y) \in f\}.$$

Every function has an inverse relation. The inverse relation is not generally a function, as the following example shows. It is also different from the preimage operation $T \mapsto f^{-1}[T]$ introduced in Section 5.1.1.

Example 5.20.

Definition 5.21.
If $f : X \to Y$ is a function and its inverse relation is also a function, we say that f is *invertible* and use $f^{-1} : Y \to X$ to denote the resulting *inverse function*.

Question 5.22.
When is a function $f : X \to Y$ invertible?

A real-valued function that passes the "horizontal line test" has an inverse function defined on its range. For a function $f : \mathbb{R} \to \mathbb{R}$ to have an inverse function whose domain is all of $\mathbb{R}$, it must also be surjective. One way to understand the horizontal line test is that the graph of an inverse relation is the reflection of the original graph through the line $y = x$. Since we want this reflection to pass the vertical line test, no horizontal line should meet the original graph more than once. This is the injectivity part of invertibility.

Thus the horizontal-line test captures injectivity, but invertibility also depends on the stated codomain. We begin by proving that every invertible function is injective.

Theorem 5.23.
If $f : X \to Y$ is an invertible function, then f is injective.

Proof.

Unfortunately, this is not a complete answer to our question because the converse of Theorem 5.23 is not true. We have found a condition that all invertible functions must satisfy, but not all functions that satisfy this condition are invertible. The following example illustrates the difficulty.

Example 5.24.

For a function $f : X \to Y$ to be invertible, every element of the codomain Y must be a first coordinate of some ordered pair in the inverse relation. That in turn means that every element of the codomain must be the second coordinate of some ordered pair in f. This is exactly what it means to say that f is surjective. This leads us to believe the following:

Theorem 5.25.
If $f : X \to Y$ is invertible, then f is surjective.

Proof.

We are now ready to give a complete answer to our question in the form of the following theorem.

Theorem 5.26.
A function $f : X \to Y$ is invertible if and only if it is bijective.

Proof.

Chapter 6

The Real Numbers

Introduction

We now turn our attention to developing a mathematical description of the real numbers. All of the results of calculus can be derived from these basic properties of the real number system, which is usually done in a first course in real analysis. Our basic assumptions about the real numbers are called axioms, and they are divided into several groups. Many of these axioms will be familiar to you from previous courses in algebra.

6.1 Field Axioms

Taken as a group, the field axioms tell us that the real numbers together with the operations of addition and multiplication form what is known as a *field*. Very informally, you might think of a field as something that satisfies the usual rules you encountered in high school algebra. Fields and related objects are studied in more depth in a course in abstract algebra.

Field Axioms.
The set of real numbers $\mathbb{R}$ is a collection of objects together with the two binary operations addition and multiplication that satisfy the following properties (known as the field axioms):

(i) Commutative Laws: For every $a, b \in \mathbb{R}$, $a + b = b + a$ and $ab = ba$.

(ii) Associative Laws: For every $a, b, c \in \mathbb{R}$, $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$.

(iii) Distributive Law: For every $a, b, c \in \mathbb{R}$, $a(b + c) = ab + ac$.

(iv) Identities: There are distinct elements 0 and 1 in $\mathbb{R}$ such that $a \cdot 1 = a$ and $a + 0 = a$ for every $a \in \mathbb{R}$.

(v) Additive Inverses: For every $a \in \mathbb{R}$, there is an element $-a \in \mathbb{R}$ such that $a + (-a) = 0$.

(vi) Multiplicative Inverses: For every $a \in \mathbb{R}$ with $a \neq 0$, there is an element $a^{-1} \in \mathbb{R}$ such that $aa^{-1} = 1$.

Any set of objects satisfying all of these properties is a *field*. The rational numbers $\mathbb{Q}$ and complex numbers $\mathbb{C}$ are also fields. The additional order and completeness axioms below distinguish $\mathbb{R}$ from these examples.

A number of the familiar algebraic properties of the real numbers are immediate consequences of the field axioms. A few of them are collected in the following theorem, though there are certainly many others.

Theorem 6.1.
For any real numbers a, b, c, the following are true:

(i) *If* $a + b = a + c$, *then* $b = c$.

(ii) $-(-a) = a$.

(iii) *If* $a \neq 0$ *and* $ab = ac$, *then* $b = c$.

(iv) *If* $a \neq 0$, *then* $(a^{-1})^{-1} = a$.

(v) $a \cdot 0 = 0$.

(vi) $-a = (-1)a$.

(vii) *If* $ab = 0$, *then* $a = 0$ *or* $b = 0$.

Proof.

6.2 Order Axioms

As noted previously, there are a number of common fields. One of the differences between $\mathbb{R}$ and $\mathbb{C}$ is that we think of the real numbers as lying in order along a line. There is no natural way to organize the complex numbers in this fashion. The next group of axioms make precise what we mean when we say that the real numbers form an *ordered field*.

Order Axioms. There is an order $<$ defined on the real numbers $\mathbb{R}$ satisfying:

(i) Transitivity: If $a < b$ and $b < c$, then $a < c$.

(ii) Trichotomy: For every two real numbers a and b, exactly one of the following holds:

$$a < b \quad \text{or} \quad a = b \quad \text{or} \quad b < a$$

(iii) If $a < b$, then $a + c < b + c$ for every $c \in \mathbb{R}$.

(iv) If $a < b$ and $c > 0$, then $ac < bc$.

We derive several consequences of the fact that $\mathbb{R}$ is an ordered field.

Theorem 6.2.
The following statements are true in $\mathbb{R}$*:*

(i) $a > 0$ *iff* $-a < 0$.

(ii) *If* $a < b$*, then* $-a > -b$.

(iii) *If* $a \neq 0$*, then* $a^2 > 0$.

(iv) $1 > 0$.

Proof.

Theorem 6.3.
Let $a, b \in \mathbb{R}$.

(i) If $a > 0$ and $b > 0$, then $ab > 0$.

(ii) If $a < 0$ and $b < 0$, then $ab > 0$.

(iii) If $a > 0$ and $b < 0$, then $ab < 0$.

Theorem 6.4.
Let $a \in \mathbb{R}$.

(i) If $a > 0$, then $a^{-1} > 0$.

(ii) If $a < 0$, then $a^{-1} < 0$.

Proof.

Theorem 6.5.
Let $a \geq 0$ and $b \geq 0$. Then $a < b$ iff $a^2 < b^2$.

Proof.

6.3 Completeness of the Real Numbers

Our axioms so far ensure that $\mathbb{R}$ is an ordered field. The field $\mathbb{Q}$ of rational numbers is also an ordered field, so we need something further to distinguish between $\mathbb{R}$ and $\mathbb{Q}$. Our last axiom is based on the observation that the field $\mathbb{Q}$ has *holes*, whereas $\mathbb{R}$ does not. Let's first consider an example to clarify what we mean by this.

Example 6.6.

6.3.1 Upper and Lower Bounds

Definition 6.7. Let A be a nonempty set of real numbers. We say that a number u is an *upper bound* for A if $x \leq u$ for every $x \in A$. We say that a number m is a *lower bound* for A if $x \geq m$ for every $x \in A$. We say that A is *bounded above* if A has an upper bound, that A is *bounded below* if A has a lower bound, and that A is *bounded* if A is bounded above and below.

Example 6.8.

The previous example illustrates several things. It is possible for a nonempty set to have both upper and lower bounds, just one bound, or no bounds at all. Upper and lower bounds may or may not be elements of the set. Finally, upper and lower bounds are not unique. Transitivity implies that if M is an upper bound for a set A, then every number larger than M is also an upper bound for A. Similarly, if m is a lower bound for A then every number smaller than m is a lower bound for A.

Consider the set $A = \{a \in \mathbb{R} \mid a^2 \leq 2\}$ from the previous example again. We claimed that 2 is an upper bound, and that is certainly true, but note that 3/2 is also an upper bound. In some sense this smaller upper bound is a "better" bound for A because it puts a tighter restriction on the size of the elements of A. This is approximately like saying that Grand Forks is a better way to describe the location of UND than North Dakota. In this sense the best possible upper bound would be the smallest one, if there is one. In this case, A has a smallest upper bound in $\mathbb{R}$, namely $\sqrt{2}$. The analogous rational set $\{q \in \mathbb{Q} \mid q^2 \leq 2\}$ has no least upper bound in $\mathbb{Q}$. This is the basic difference between $\mathbb{Q}$ and $\mathbb{R}$ that we wish to capture in an axiom. First, we need a couple more definitions.

Definition 6.9. Let A be a nonempty subset of $\mathbb{R}$. We say that a number u is a *least upper bound* (LUB) for A if both of the following are true:

(i) For every $a \in A$, $a \leq u$.

(ii) For every upper bound b of A, $u \leq b$.

We say that a number m is a *greatest lower bound* (GLB) for A if both of the following are true:

(i) For every $a \in A$, $a \geq m$.

(ii) If b is a lower bound for A, then $m \geq b$.

Note that a LUB is sometimes called a *supremum* and a GLB is sometimes called an *infimum*.

It is not hard to prove that, unlike upper bounds, a set can have only one least upper bound. This fact is made explicit in the following theorem, whose proof is left as a homework exercise.

Theorem 6.10.
Let A *be a nonempty subset of* $\mathbb{R}$. *If* u *and* v *are least upper bounds for* A, *then* $u = v$.

The following theorem says that the least upper bound of a set must in some sense be "close to" the set.

Theorem 6.11.
Let A *be a nonempty subset of* $\mathbb{R}$ *and let* u *be the least upper bound for* A. *If* $x < u$, *then there is an element* $a \in A$ *such that* $x < a$.

Proof.

We are now ready to state our final axiom, which says that $\mathbb{R}$ is *complete*.

The Completeness Axiom. Let A be a nonempty subset of $\mathbb{R}$. If A has an upper bound, then A has a least upper bound.

There are a number of consequences of the Completeness Axiom, most of which are beyond the scope of this text. We will look at only a couple of them. We begin with the fairly intuitive-seeming fact that for any real number x, there is a natural number larger than x.

Theorem 6.12.
(Archimedean Property) *If $x \in \mathbb{R}$, then there is an $n \in \mathbb{N}$ such that $n > x$.*

Proof.

We previously commented, without proof, that the Completeness Axiom would allow us to distinguish $\mathbb{R}$ from $\mathbb{Q}$. We will now make this explicit by proving that there is at least one real number that is not rational.[1] By Theorem 2.18, there is no rational number whose square is 2. We now show that the Completeness Axiom implies that there must be a real number x with $x^2 = 2$.

Theorem 6.13.
There is a number $x \in \mathbb{R}$ such that $x^2 = 2$.

Proof.

[1] In fact there are more irrational numbers than there are rational numbers, as we shall see in the final chapter.

Chapter 7

Introduction to Cardinality

Introduction

What do we mean when we say that there are *four* suits in a standard deck of cards? More generally, what does it mean to count any collection of objects? Since you may no longer actually think about counting, it may help to watch a child who is just learning to count. Given four objects to count, the child is likely to point to each object in turn and count "one, two, three, four." In other words, the child is explicitly constructing a bijective function between the objects she is counting and the elements of the set $\{1, 2, 3, 4\}$. Our goal in this chapter is to extend this idea to infinite sets.

7.1 The Cardinality of a Set

For sets A and B, write $A \equiv B$ if there is a bijective function $f : A \to B$. In this case we say that A and B are *equinumerous* or that they have the same *cardinality*. Write $A \preceq B$ if there is an injective function $f : A \to B$; equivalently, $B \succeq A$. We write $A \prec B$ to indicate that $A \preceq B$ and $A \not\equiv B$. It is also common to write $|A| = |B|$ rather than $A \equiv B$.

Theorem 7.1.
For any set C whose elements are sets, equinumerosity restricts to an equivalence relation on C.

Proof.

Note that Theorem 7.1 allows us to say that two sets A and B have the same cardinality if we are able to find a bijection from A to B or a bijection from B to A.

Theorem 7.2.
On any set C whose elements are sets, the relation $\preceq$ is reflexive and transitive.

Proof.

While we would not expect $\preceq$ to be symmetric, we can ask for a cardinal analogue of antisymmetry. Mutual injections do not force two sets to be literally equal, but do they force the sets to be equinumerous? Georg Cantor (1845–1918) was interested in this question, and Felix Bernstein (1878–1956) proved that the answer is yes. The result is usually known as the Cantor–Bernstein Theorem.

Theorem 7.3 (Cantor–Bernstein).
If $A \preceq B$ *and* $B \preceq A$, *then* $A \equiv B$.

We prove this theorem in Chapter A. The following example uses the same kind of chain-shifting idea to construct a bijection between the open interval $(-1, 1)$ and the closed interval $[-1, 1]$.

Example 7.4.

7.2 Finite Sets

Question.
What does it mean to say that a set A is finite?

At first glance, this may seem like something you've known for a long time. Don't we just mean that we can count the elements of A? If so, is the number of cells in your body finite? Can you count them? Maybe the answer to our question isn't quite so obvious after all.

For each $n \in \mathbb{N} \cup \{0\}$, define

$$I_0 = \varnothing \qquad \text{and} \qquad I_n = \{1, 2, \dots, n\} \quad \text{when } n \in \mathbb{N}.$$

Thus $I_4 = \{1, 2, 3, 4\}$. These sets let us formalize the idea of counting while handling the empty set without a separate convention.

Definition 7.5.
Let A be a set and let $n \in \mathbb{N} \cup \{0\}$. We say that A has n elements if $A \equiv I_n$. The set A is *finite* if $A \equiv I_n$ for some $n \in \mathbb{N} \cup \{0\}$, and it is *infinite* if it is not finite.

Theorem 7.6. *The set $\mathbb{N}$ is infinite.*

Proof.

Composing a bijection $A \to I_n$ with the inclusion $I_n \hookrightarrow \mathbb{N}$ gives the following.

Theorem 7.7.
If A is finite, then $A \preceq \mathbb{N}$.

Now that we have a definition of finite, let's reconsider the set C of cells in your body. Is C finite? If so, for which n is $C \equiv I_n$? We still can't really count them, so perhaps we've just obscured the question rather than answering it. Scientists estimate that there are about 10,000,000,000,000 cells in the average adult human body. That's not an actual count of the number of cells in any individual human body, though. These kinds of estimates are based on the sizes of various kinds of cells and the approximate proportion of each kind of cell in the body. In fact, we could determine the maximum number of cells that might be in a person's body by figuring out how many of the smallest kinds of cells would be required to build a body of a particular volume, or weight, etc. It seems reasonable to think that we could say a set was finite if we were sure it had at most n elements for some natural number n. That is the intent of the next result.

Theorem 7.8.
If $\mathrm{A} \preceq \mathrm{I}_n$ *for some* $n \in \mathbb{N} \cup \{0\}$, *then* A *is finite.*

Suppose there is an injection $f : \mathrm{A} \to \mathrm{I}_n$. Our definition of finite asks for a bijection with some I_k, while f need not be surjective. The next lemma removes one unused point from the codomain at a time.

Lemma 7.9.
Let $n \in \mathbb{N}$. *If* $f : \mathrm{A} \to \mathrm{I}_n$ *is injective but not surjective, then there is an injection* $g : \mathrm{A} \to \mathrm{I}_{n-1}$.

Proof.

We will now prove the theorem.

Proof of Theorem 7.8.

Consequently, if $\mathbb{N} \preceq A$, then A is infinite.

We conclude this section with a question, the answer to which may seem obvious to you.

Question 7.10.
If $m, n \in \mathbb{N} \cup \{0\}$ and $m \neq n$, can you prove that $\mathrm{I}_m \not\equiv \mathrm{I}_n$?

7.3 Denumerable Sets

Georg Cantor was able to define a complete system of infinite numbers and of arithmetic on those numbers. We will not discuss his system here, but we will look at one particular kind of infinite number that is important in many areas of mathematics.

Theorem 7.11.
If A is an infinite set and B is a finite subset of A, then the set $A \setminus B$ is infinite.

Proof.

Definition 7.12.
Let A be a set. We say that A is:

- *denumerable* if $A \equiv \mathbb{N}$.
- *countable* if A is either finite or denumerable.
- *uncountable* if A is infinite and not denumerable.

Example 7.13.

The preceding example points out an important difference between finite and infinite sets. The set A is a proper subset of $\mathbb{N}$, but has the same cardinality as $\mathbb{N}$. A set equinumerous with one of its proper subsets is called *Dedekind-infinite*. To see why a finite set cannot have this property, suppose $A \equiv I_n$ and $B \subsetneq A$. The image of B under a bijection $A \to I_n$ is a proper subset S of I_n; in particular, $n \geq 1$. The inclusion $S \to I_n$ is not surjective, so Lemma 7.9 gives an injection $S \to I_{n-1}$. If $B \equiv A$, this would produce an injection $I_n \to I_{n-1}$, contradicting the induction in the completion of Question 7.10. Therefore every Dedekind-infinite set is infinite. The converse holds in standard ZFC set theory, but its proof uses a form of the Axiom of Choice; without Choice, the two notions need not coincide.

We would like to determine which of our results about finite sets are also true for denumerable sets. If A and B are denumerable, must $A \cup B$ also be denumerable? Are subsets of denumerable sets denumerable? Is there an analog of Theorem 7.8? Are there other ways to tell that a set is denumerable?

Theorem 7.14.
If A *is a denumerable set and* $B \equiv A$, *then* B *is denumerable.*

Proof. By definition we have $A \equiv \mathbb{N}$. Since $\equiv$ is an equivalence relation, it follows that $B \equiv \mathbb{N}$ as desired.

Theorem 7.15.
If $A \subseteq \mathbb{N}$*, then* A *is countable.*

Proof.

Corollary 7.16. *A set* A *is countable if and only if* $A \preceq \mathbb{N}$.

Proof.

Corollary 7.17. *A subset of a countable set is countable.*

Proof.

Thus a set A is countable exactly when there is an injection from A to $\mathbb{N}$. If A is nonempty, this is also equivalent to finding a surjection from $\mathbb{N}$ onto A. In one direction, invert an injection on its range and send unused natural numbers to a fixed element of A; in the other, assign each $a \in A$ the least natural number mapping to it. The nonempty hypothesis is necessary: there is no function from $\mathbb{N}$ onto $\varnothing$.

Corollary 7.18.
A set A *is countable if either of the following is true:*

(i) *There is an injection* $f : A \to \mathbb{N}$.

(ii) $A \neq \emptyset$ *and there is a surjection* $f : \mathbb{N} \to A$.

Theorem 7.19.
The union of two denumerable sets is denumerable.

Proof.

Using Mathematical Induction on the number of sets, we obtain:

Corollary 7.20.
The union of any positive finite number of denumerable sets is denumerable.

Corollary 7.21. *The union of two countable sets is countable.*

Proof.

Theorem 7.22.
The set $\mathbb{N} \times \mathbb{N}$ *is denumerable.*

Proof.

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7.3.1 The Rational Numbers

At first glance it may seem that there are more rational numbers than there are natural numbers. After all, there are infinitely many rational numbers between any two natural numbers. One of Cantor's accomplishments was to show that the set of rational numbers is actually denumerable. Our intuition developed from years of working with finite sets just doesn't serve us very well when working with infinite sets.

Lemma 7.23.
The set $\mathbb{Q}^+$ of positive rational numbers is countable.

Proof.

Since the function taking every positive rational to its additive inverse is bijective, the following corollary is immediate from the lemma.

Corollary 7.24.
The set $\mathbb{Q}^-$ of negative rational numbers is countable.

We know that $\mathbb{Q}^+$, $\mathbb{Q}^-$, and $\{0\}$ are all countable sets. Applying Corollary 7.21 twice, we conclude:

Theorem 7.25.
The set $\mathbb{Q}$ of rational numbers is denumerable.

Proof.

7.3.2 The Real Numbers

By this point it may seem possible that all sets are countable, making denumerability a useless distinction. We will show that this is not true by proving that the real numbers are uncountable. A real number can have two decimal expansions; for example, $1.000\ldots = 0.999\ldots$. To make decimal digits unambiguous, we choose for each real number its *canonical* expansion: the one that is not eventually all 9s.

Theorem 7.26.
The set $\mathbb{R}$ of real numbers is uncountable.

Proof.

Appendix A

The Cantor–Bernstein Theorem

The relation $X \preceq Y$ means that X injects into Y. It is reflexive and transitive, but it is not literally antisymmetric on sets: distinct sets can have the same cardinality. The Cantor–Bernstein Theorem supplies the appropriate cardinal version of antisymmetry:

$$X \preceq Y \text{ and } Y \preceq X \quad \Longrightarrow \quad X \equiv Y.$$

The proof below is constructive once the two injections are given.

A.1 The Forward-Chain Construction

Assume that $f : X \to Y$ and $g : Y \to X$ are injective. Some points of X have no predecessor under g; begin a forward chain at each such point. Define

$$A_0 = X \setminus g[Y], \qquad A_{n+1} = g\big[f[A_n]\big] \quad (n = 0, 1, 2, \ldots),$$

and let

$$A = \bigcup_{n=0}^{\infty} A_n = \{x \in X \mid x \in A_n \text{ for some } n \in \mathbb{N} \cup \{0\}\}.$$

On the chains in A we use f in the forward direction. Off those chains, every point has a unique predecessor under g, so we use that predecessor.

Proof of Theorem 7.3.

A.2 Why the Construction Works

The set A_0 records where the backward search through g must stop. Each A_{n+1} advances those starting points one step through f and then g. Using f on every such forward chain prevents a missing predecessor from causing a gap. Everywhere else, the g-chains extend backward indefinitely, so reversing g is safe. The argument also covers empty sets without a separate case.